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Accurate multi-category student performance forecasting at early stages of online education using neural networks

Naveed Ur Rehman Junejo[1,2], Muhammad Wasim Nawaz[3], Qingsheng Huang[4]✉, Xiaoqing Dong[1], Chang Wang[1] & Gengzhong Zheng[2]✉

The ability to accurately predict and analyze student performance in online education, both at the outset and throughout the semester, is vital. Most of the published studies focus on binary classification (Fail or Pass) but there is still a significant research shortcoming in predicting performance of students across multiple categories. This study introduces a novel neural network-based approach capable of accurately predicting student performance and identifying vulnerable students at early stages of the online courses. The open university learning analytics (OULA) dataset is employed to develop and test the proposed model, which predicts outcomes in Distinction, Fail, Pass, and Withdrawn categories. The OULA dataset is preprocessed to extract features from demographic data, assessment data, and clickstream interactions within a virtual learning environment (VLE). Novel features engineering has been utilized to predict students' performance across multiple categories at early stages of courses. Specially, students' VLE interactions are aggregated by total clicks to represent daily engagement and assess online activity. Comparative simulations indicate that the proposed model significantly outperforms existing baseline models including artificial neural network long short-term memory (ANN-LSTM), random forest (RF) 'gini', RF 'entropy' and deep feed forward neural network (DFFNN) in terms of accuracy, precision, recall, and F1-score. The results indicate that the prediction accuracy of the proposed method is about 25% more than the existing state-of-the-art methods. Furthermore, compared to existing methodologies, the model demonstrates superior predictive capability across temporal course progression, achieving superior accuracy even at the initial 20% phase of course completion.

Keywords Early-prediction, Machine Learning, Deep Learning, Feature Engineering, Virtual Learning Environment

Predicting and understanding student achievement is one of the biggest challenges faced by educators. Online learning environments continue to struggle with the issue of high dropout rates, making accurate predictions crucial for reducing this rate[1–5]. The primary goal of the current work is to develop a predictive model to identify students who are at risk of dropping out early. This prediction model will be extremely helpful to instructors, enabling them to make timely and effective interventions, which will reduce the dropout rate. Massive open online courses (MOOCs) and other online education platforms have grown in popularity recently, offering students more opportunities to acquire learning materials and advance their careers. However, a significant concern is the high dropout rate associated with these online courses[1,6,7]. MOOCs have become increasingly important in education because they provide students with access to resources that help them learn and develop their skills.

Even with the advantages of virtual learning environments (VLEs), high dropout rates remain a problem. It makes sense that, to reduce dropout rates and increase course continuity, educators and academics are highly interested in developing models and tactics that can assess students' behavior and personal data. Identifying

[1]School of Physics and Electronic Engineering, Hanshan Normal University, Chaozhou 521041, China. [2]Department of Computer Science and Engineering, Hanshan Normal University, Chaozhou 521041, China. [3]Department of Computer Engineering, The University of Lahore, Lahore, Punjab 54000, Pakistan. [4]School of Mathematics and Statistics, Hanshan Normal University, Chaozhou 521041, China. ✉email: huangqs@hstc.edu.cn; zhenggz@hstc.edu.cn

at-risk students who may be likely to drop out can be challenging using traditional educational methodologies, making it difficult to offer effective interventions[8,9]. Deep feed-forward neural network (DFFNN)[4], sequential conditional generative adversarial network (SC-GAN)[5], artificial neural network (ANN)[10], joint deep neural network namely recurrent neural network-gated recurrent unit (RNN-GRU)[11], reduced training vector-based support vector machine (RTV-SVM)[12], supervised machine learning (ML) models such as support vector machine (SVM), random forest (RF), and decision tree (DT) using the expectation maximum technique[13], long short-term memory (LSTM)[14], and ANN-LSTM[15]have been proposed by researchers to anticipate students who are at risk of dropping out in an attempt to address this issue. However, these existing approaches can be tedious and time-consuming and often depend on manually created features. To address these issues, recent research has suggested building predictive models for student performance and dropout prediction using ML and deep learning (DL) algorithms[16,17].

Achieving analytics objectives is made easier by understanding the performance of a class of students during the first few days of a course in VLE. Numerous academics have suggested predictive models to forecast student success in MOOC in binary classifications (dropout/not) or (pass/fail)[10,11,18–23]. Nevertheless, a limited number of studies have created models that forecast the performance of the students in multiclass categories, such as[4,15,24–27]. Therefore, further research is required to enhance the prediction performance of multi-class models[15,28]. While more sophisticated DL techniques are rarely used in comparison with traditional artificial intelligence techniques which are frequently employed in predicting student success in online higher education[29]. Since most research has focused on creating prediction models for specific courses, such as[13,30,31], overfitting may occur if new courses are created[32]. Figure 1 represents the difference between binary and multiclass classification on dataset. Binary classification 1 indicates that in literature the pass and distinction categories fall under the 'Pass' category whereas the category 'Fail' is assigned to withdrawn and fail candidates. Binary classification 2 shows that researchers only considered pass and fail data where distinction and withdrawn information have been ignored. Multiclass classification indicates that all four categories contain distinct information and should be grouped separately as Fail, Pass, Distinction, and Withdrawn. In this study, 'Distinction' refers to students who achieve outstanding academic performance, typically ranking in the highest percentile of their course. On the other hand, 'Withdrawn' refers to students who leave the course before completion, either voluntarily or due to institutional policies. Understanding these categories is crucial, as early identification of at-risk students (such as those likely to withdraw) allows for timely interventions, while predicting distinction-level students can help in talent recognition and resource allocation.

Therefore, it is important to develop predictive models to evaluate various courses to address this issue. The lack of study on multiclass classification with the OULA dataset is due to difficulties to differentiate multiple classes. Current approaches have limitations to capture complex correlations between various variables from different dataset files, which are essential for precise multiclass predictions. It is critical to close this gap because multiclass categorization offers a more practical picture of student performance. This enables individualized support and focused interventions. Our work closes this gap by improving the integration and distillation of information from multiple datasets and by developing a novel data preprocessing pipeline for feature engineering.

Although the proposed method is tested on the OULA dataset, it is possible to generalize the proposed method to other online learning platforms. The dataset's features, including demographic attributes, assessment performance, and clickstream activity are commonly recorded in most Learning Management Systems (LMS) such as Moodle, Canvas, and Coursera. Thus, the proposed feature engineering approach, including clickstream

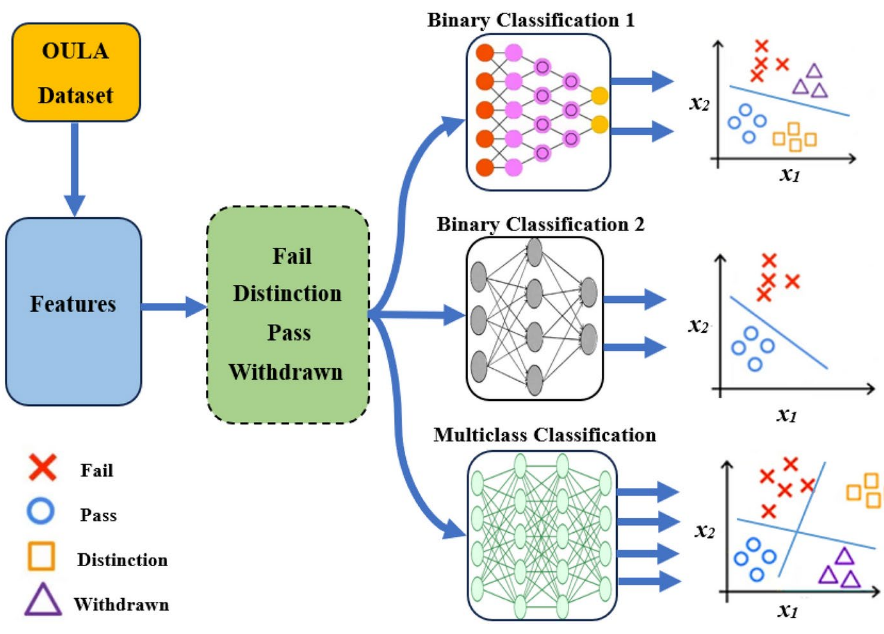

Fig. 1. Binary and multiclass classifications for student performance evaluation.

aggregation, can be adapted for different LMS platforms with minimal modifications. Furthermore, the model is designed to handle diverse course structures, which makes it applicable to other similar datasets. The main contributions of this work include:

- We propose a novel data preprocessing pipeline that effectively merges and aggregates multiple dataset files, retaining crucial features.
- Our method integrates various dataset files to provide a unified view of student data and to enable accurate predictions through novel features.
- Our approach introduces innovative feature engineering techniques that distil data from multiple dataset files into informative attributes, enhancing the accuracy of early student performance prediction.
- We present and utilize a one-dimensional convolutional neural network (1D-CNN) model for multiple categories that achieves outstanding accuracy, precision, recall, and F1-score in predicting student performance in online learning environments. Furthermore, we also analyze our method in terms of the area under the curve (AUC), where the high AUC values reflect our proposed method's ability to distinguish between different levels of students' achievement.
- By accurately early predicting student performance at about 20% of the course length, our approach facilitates timely interventions to improve learning outcomes in online education environments, our proposed model attains 92% of accuracy.
- The results from experimental data reveal that the proposed model tops the existing, RF 'entropy', RF 'gini, ANN-LSTM, and DFFNN, models regarding precision and recall of 98%.

The rest of this paper is structured as follows. Section 2 presents the comprehensive literature review for performance prediction of the students. The explanation of the proposed methodology, which contains the detailed description of the dataset, data preprocessing, and the proposed 1D-CNN model is mentioned in Section 3. Section 4 analyzes the experimental results and provides detailed discussions on our study. The conclusion is given in Section. 5 followed by research highlights and proposals for future research.

Literature review and background study

Wasif et al. provided a way to use logging data from the OULA dataset to predict student performance in their 2019 study. For comparison in binary classification, they used multilayer perceptron (MLP), logistic regression (LR), Gaussian Naïve Bayes, and RF models. The findings showed that the RF model performed better than the others, with 88%, 88%, 89%, and 89%, for F1-score, recall, precision, and accuracy, respectively. The models' ability to accurately identify children who are at risk allows for prompt interventions that improve student progress, according to the authors' conclusion[18]. In 2019, Hassan et al. proposed a model named the LSTM deep model to predict withdrawn students using only clickstream data of OULA dataset. A binary classification problem has been considered where the "pass" class has been merged with the "distinction" to formulate one class and the other was the "withdrawn" class. The accuracy was approximately 80% in the fifth week and approximately 97% in the twenty-fifth. The proposed model namely LSTM outperforms LR and artificial neural networks in terms of accuracy, precision, and recall[19]. In 2019, Aljohani et al. used a DL model to identify students who were likely to fail a course, allowing instructors to provide the appropriate support on the clickstream data. The suggested model has a 90% accuracy rate in predicting pass and fail grades in the first 10 weeks of student engagement in a VLE. In the pass and fail classification, the LSTM model superseded the baseline LR and ANN in terms of precision and recall value by 93.46% and 75.79%, respectively[20].

In 2019, Waheed et al. proposed an ANN DL based model for predicting the students' at-risk whether they are distinct, withdrawn, and fail by utilizing the click-stream and demographic data. The problem has been considered as a binary categorization for the final result. Out of a total of 54 features, the authors employed sparse reduction to choose 30 features, and then the "MinMax" scaler is employed to transform the values of certain features. The fail predictive model shows that the information of withdrawn students has been neglected and the metrics of distinction and pass students have been combined. This model provides accuracy of almost 77%, 81%, 86%, and 88% in the quarter1 to quarter4 i.e., Q1, Q2, Q3 and Q4, respectively. Fail data was overlooked, and distinction and pass data have been merged into one class in the scenario of withdrawn prediction. The withdrawn predictive model attained an accuracy of about 78% in Q1, 86% in Q2, 90% in Q3, and 93%in Q4. ANN model results attained a better performance in comparison of the baseline LR and SVM models in terms of accuracy[10]. He et al., in 2020, suggested the joint deep neural network namely RNN-GRU adequate for both the sequential and static data. It also implemented the data completion mechanism to fill the missing data stream. The author performed a binary classification to predict outcomes of online course where the 'pass' and 'distinction' categories were considered instead of the 'pass' and 'withdrawal' classes. Results revealed that methods such as RNN and GRU outperformed LSTM. Therefore, the RNN-GRU model reached over 80% accuracy to predict the 'at risk' students at the end of the semester. The forecast applied to the entire course. Thus, the most recent courses served as a test set. Throughout all courses, the average accuracy from the fifth to the thirty-ninth week ranged from 60% to 90%[11].

Based on academic performance to forecast at-risk or marginal pupils, Chui et al. (2020) developed a RTV-SVM classifier. 93.5% and 93.8%accuracy has been attained by TRV-SVM for marginal and at-risk students, respectively[12]. In order to anticipate students' final exam marks in 2021, Kumar et al. assessed the effectiveness of classification models such as RF, NB, SVM, K-NN, and ANN utilizing clickstream, assessment, and demographic data as input of the prediction algorithms[21]. Esteban (2021) investigated the significance of assignment data for predicting students' performance. A multiple-instance learning prediction algorithm was suggested by the authors to predict which students will pass and fail. The model's input has been the assessment data[23]. In 2021, Hlioui et al. used assessment, clickstream, and demographic predict withdrawal rates in students using a variety of models, including RF, DT (J48), MLP, SVM, and Bayesian (TAN) classifiers. Two classes of performance prediction have been identified: withdrawal and completion (sometimes known as "Distinction", "Pass", or "Fail")[22].

In 2021, Adnan et al. employed a DFFNN to estimate the final grading of students such as pass, distinct, fail, and withdrawn. The greatest average accuracy recorded at the end of the course was 43%, 63%, 71%, and 72% when the input data consisted of demographic information, demographic information and click steam, demographic information, click stream, and assessment, and all features, respectively. By the end of the courses, 90%accuracy had been achieved for the final grades while considering the binary classification[4]. An ANN semi-supervised learning ensemble model was proposed by Vo et al. in 2021. It estimated the performance of the students prior to the midterm exam and at the conclusion of the course i.e., pass or fail. Five different courses including AAA, BBB, CCC, DDD, and FFF have been considered, and these courses were selected to be inputs into the proposed model. Every selected subject was offered in two distinct semesters. While the second course was used for testing, the first one was used for training. Averaging 87.47%, the chosen courses' accuracy during the midterms of the semester. About $40\%$of the OULA dataset student data were disregarded when it came to distinction and withdrawal. The proposed ANN model has been compared to baselines KNN, RF, NB, C4.5, Logistic, and SVM[30]. Pei et al, in 2021, used the expectation maximum technique to develop supervised ML models including DT, RF and SVM. This model predicted about the drop-out/stay of the student the preceding week. Moreover, to improve the accuracy of prediction for the running week, the resultant probability is applied as input. The aggregated clickstream and demographic data of two STEM and social sciences courses i.e., CCC-2014 J and CCC-2014B, and AAA-2014 J and AAA2013, respectively is used as input. The data is balanced by applying the Synthetic Minority Over-Sampling (SMOTE) technique. The average accuracy of almost $88\%$is achieved during all the weeks for the selected courses[13].

A DL-based approach i.e., SC-GAN implemented by Waheed et al. in 2021 assisted in generating the synthetic records of student for the next timestamp by capturing past performance of each student for its former sequences. The authors included the "fail" and the updated "pass" classes in their binary classification to predict the students' learning performance. The experimental results validated that SC-GAN outperformed Synthetic Minority Oversampling and conventional Random Over-sampling in terms of AUC of 6.53% and 7.07%, respectively[5]. In 2022, Qiu et al. presented a prediction framework based on six classic ML algorithms including SVC (R), SVC (L), Naïve Bayes (NB), K-Nearest Neighbor (KNN), and softmax. The predictor's projected output value was either qualified or unqualified. In 12 activities, the authors employed click stream data for "DDD" course data[33].

In 2022, Adnan et al. considered a multiclass classification scenario with four classes including pass, distinct, withdraw, and fail to predict student performance. Authors implemented DL and ML models such as DFFNN, RF with two different criteria 'gini' and 'entropy', SVM, DT, KNN, LR, AdaBoost, Gradient Boosting, Bernoulli Naïve Bayes, Gaussian Naïve Bayes, and Extra Tree classifier to compare their performance. Experimental results witnessed that the RF 'gini' and RF 'entropy' outperformed in terms of accuracy as compared other models[24]. When employing clickstream and demographic data, multi-class student performance prediction models still require to improve their accuracy early in MOOC courses. Yanqing Xie in 2021 a proposed model named; attention-based multi-layer LSTM (AML) to predict student performance, which combines student demographic data and clickstream data as input of the model for comprehensive analysis[34]. Although, the ML and DL based predictive models proposed by Adnan (2021)[4]and (2022)[24] are multiclass classification, these methods were not used early in the course period, the analysis of models in term of accuracy attained at the end of the course were 63% and 66%, respectively, which can be considered low results[34]. Yanqing Xie has considered two scenarios; one is four-class classification and the other is binary classification. The accuracy obtained for four-class classifications from week 0, week 5, and week 25 were 43.93%, 53.51%, and 57.40%, respectively. The accuracy attained for binary classification from week 0, week 5, and week 25 were 65.62%, 78.94%, and 95.75%, respectively. Experimental results have proven that the AML model attained better accuracy then state-of-the-art models for binary as well as four-class classifications[34].

As a multiclass classification model, Hao Jia et al. in 2022 proposed a students' performance prediction network (SPBN) model to predict the performance of the students in the final, only assessment data has been considered as input. In addition, researchers compared it with different classical ML models. Based on the students' assessment results, the computed performance value was split into four categories: fail, pass, good, and distinction. SPBN has attained almost 86.6% average F1-score, while the same metrics are about 81.4%, 80.90%, 74.9%, 85.5%, 71.3%, and $55.2\%$for LightGBM, XGBoost, AdaBoost, RF, MLP and Naïve Bayes, respectively[25]. In 2023, Waheed et al. presented the results of a study that examined how well-known DL technology LSTM performed in predicting which students would fail a course that was given in an online, self-paced learning environment. Surprisingly, the LSTM algorithm achieves up to 71%accuracy with only the first five weeks of course activity log data used for training to distinguish between pass and fail classes. This surpasses nearly all other traditional algorithms, even though the LSTM algorithm is trained on the entire dataset collected for the course, i.e., up to 38 weeks[14]. In 2023, Nafea proposed an approach using group modeling to identify the students' performance. Authors have merged the strengths of ML algorithms namely DT, SVM, AdaBoots, and RF. In this study, the 'pass' and 'distinct' labels were merged and allotted as the 'pass' label. Similarly, the 'fail' and 'withdrawn'labels were merged, and allocated the 'fail' label. The results have proven that the proposed idea outperforms the baseline by attaining 95%accuracy[35].

Fatima Ahmed Al-Azazi in 2023 proposed a DL based ANN-LSTM multiclass model to predict students' performance[15]. A comparative study between the two baseline methods i.e., GRU and RNN, and the proposed ANN-LSTM model is provided. The ANN-LSTM showed 70% accuracy at the end of the third month of the semester was while 53%, and $57\%$was achieved by RNN and GRU, respectively. Results ensured that ANN-LSTM is more accurate in comparison to the baseline models. Furthermore, authors also compared ANN-LSTM with DFFNN proposed by Adnan in 2021[4]. The RF 'entropy' and RF 'gini' ML models proposed by Adnan in 2022[24], and the AML model suggested by Xie in 2021[34] executed multiclass classification closely associated to the research work of the authors, hence, they were designated as the baseline models. An accuracy of almost 43% and 72%was achieved on the first and 270 th day (last day) of the course, respectively, through the testing of data[15]. Table 1 provides the summary of comparative analysis of different ML and DL methods using OULA dataset.

The existing literature on student performance prediction mainly focuses on binary classification (Pass/Fail), overlooking the nuanced distinctions between categories such as Distinction, Withdrawn, and Pass. This study bridges this gap by introducing a multiclass predictive model that provides deeper insights for early interventions. A key challenge in early-stage prediction (within the first 5–10% of the course) is the scarcity of available data, limiting the effectiveness of traditional models like ANN-LSTM or RF. By leveraging 1D-CNN, our model effectively extracts meaningful patterns from sparse early data, demonstrating superior performance in identifying student outcomes. Additionally, given the imbalance in many educational datasets, where underrepresented classes like Withdrawn are often overlooked, the proposed model incorporates class weighting to mitigate bias towards the majority class.

Methodology

In this section, we present the VLE dataset details, its pre-processing, and building and training of the neural network model. Figure 2 depicts the block diagram of our experimental workflow.

Authors	Features	Techniques applied	Objectives	Evaluation metrics	Outcome
Adnan et al.[4]	All features	DFFNN, RF, SVM, K-NN, ET, AdaBoost, Gradient Boosting, ANN	To estimate the final grading of students	Accuracy	Withdrawn, Fail, Pass, Distinction
Waheed et al.[5]	Demographic, Clickstream	SCGAN	Behavior of each student for its previous sequences; generates synthetic student records	Accuracy, F1-score, AUC	Pass, Fail
Waheed et al.[10]	Demographic, Clickstream	ANN	Merge Distinction and Pass	Accuracy, Loss, Precision, Recall	Pass/Fail, Distinction/Pass, Distinction/Fail, Withdrawn/Pass
He et al.[11]	Demographic, Assessment, Clickstream	RNN GRU	Ignore withdrawn students	Accuracy, Precision, Recall	Pass, Fail
Chui et al.[12]	Demographic, Assessment, Clickstream	RTV-SVM	Predict at-risk and marginal students	Sensitivity, Specificity, Accuracy	Pass 1 (Distinction), Pass 2, Pass 3, Pass 4, Bare Fail, Fail, Bad Fail
Pei et al.[13]	Demographic, Clickstream of specific courses	RF, SVM, DT	Limited to specific courses	Recall, Precision	Dropout (Withdrawn) or Not
Waheed et al.[14]	Demographic, Clickstream	LSTM	Students' prediction at risk of failure	Accuracy, Precision, Recall	Pass, Fail
Al-Azazi and Ghurab[15]	Demographic, Clickstream	ANN-LSTM	Predict students' performance class in VLEs	Accuracy	Distinction, Pass, Fail, Withdrawn
Wasif et al.[18]	Demographic, VLE portal	Naïve Bayes, LR, SVM, RF	Identify students at high risk of unsatisfactory performance	Accuracy, Loss, Precision, Recall	Pass, Fail
Hassan et al.[19]	Clickstream	LSTM	Identify early withdrawal; developed a week-wise sequential vector from online interactions	Accuracy, Loss, Precision, Recall	Withdrawn, Pass
Aljohani et al.[20]	Clickstream	LSTM	Ignore withdrawal students; ignore repeated courses	Accuracy	Pass, Fail
Kumar et al.[21]	Assessment, Demographic, Clickstream	NB, RF, KNN, SVM, ANN	Predict final exam scores	Accuracy, F1-score, J-Index	Pass, Fail
Hlioui et al.[22]	Assessment, Demographic, Clickstream	J48, RF, SVM, ANN, BN	Withdrawal prediction	F1-score	Withdrawal, Completion
Esteban et al.[23]	Assessment	Multiple Instance Learning	Predict students' performance using assignment data	Accuracy	Pass, Fail
Adnan et al.[24]	Clickstream, performance in past assessments, module interest, online content accessed	DT, DFFNN	Forecast students' performance in VLE by considering feature weights	F1-score, Recall, Precision, Accuracy	Withdrawn, Fail, Pass, Distinction
Hao Jia et al.[25]	Demographics, register duration, sum of clicks, average intermediate test scores	SPBN	Calculate classes based on assessment scores	F1-score	Fail, Pass, Good, Distinction
Vo et al.[30]	Assessment features of students in specific courses	Semi-supervised ANN, C4.5, KNN, Logistic NB, RF, SVM	Course-level prediction	Accuracy, F1-score, AUC	Pass, Fail
Qiu et al.[33]	Clickstreams in 12 activities for students in "DDD" course	SVC (L), NB, KNN, Softmax	Student learning performance prediction	Accuracy, F1-score, Kappa	Qualified, Unqualified
Xie[34]	Demographic and clickstream	AML	Estimated the end semester results of students	Accuracy, Precision, Recall, F1-score	Withdrawn, Fail, Pass, Distinction

Table 1. Summary of Existing Works on Student Performance Prediction.

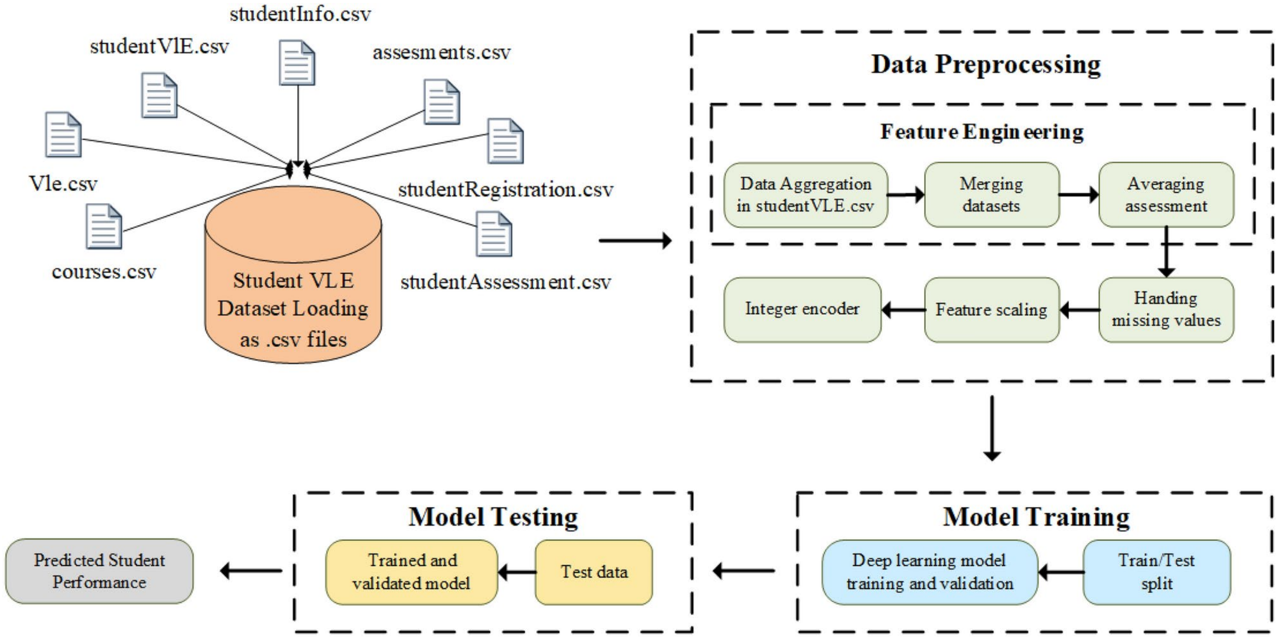

Fig. 2. Block diagram of the proposed framework.

Dataset

For training ML models, we employed an OULA dataset, which contains details about the students' demographics, course registration, interactions on the VLE, assessments, clickstreams, and final performance scores[36]. The OULA dataset is free, publicly available, and officially verified by the Open Data Institute (ODI): http://theodi.org/. In the obtained OULA dataset, the clickstream, course information, student details, and registration data are organized in seven tables, with the students serving as the main informational focus. This dataset contains data from 32,593 students collected over nine months. Four classes/grades are created from the students' final performance including Fail, Distinction, Pass, and Withdrawn. Students had access to seven courses or modules, which were offered at least twice a year at different times. Seven tables comprising the obtained raw data are linked together by identifying columns. The tables include data on assessments, types of assessments, due dates for submissions, courses, types of VLE materials, click-streams showing how students engage with the VLE, course registration details, and student demographics. The VLE clickstream contains details on the different activities that students do on heterogeneous modules, like using discussion boards, getting course materials, logging in, and visiting the main page, subpage, URL, and glossary of OU, among other things. The students' level of interest in each sort of activity is recorded by their clickstream activity.

The OULA dataset, widely used in educational data mining, contains information on student demographics, assessments, and VLE interactions from over 30,000 students. While valuable for analyzing online learning patterns, it has limitations that impact model generalization. The OULA dataset is from a single institution; thus, it has the limitation of not representing various educational settings. Moreover, another potential bias could be because online clickstream engagement patterns of this dataset could be different from other online educational platforms. Another challenge in the OULA dataset arises because of class imbalance because there are fewer instances of some classes compared to other classes. To address this issue, we use class weighting. Overfitting also poses a risk, which we try to mitigate using dropout and early stopping.

Data pre-processing

This section outlines the steps taken to pre-process OULA dataset and perform feature engineering to early predict student performance. After loading the dataset, we aggregate and merge various dataset files. The VLE data file contains information about students' interactions with the online learning platform, such as the number of clicks on course materials. These interactions are indicative of students' engagement levels.

Feature engineering

To create the new features from existing data, novel feature engineering has been proposed to improve the model's prediction power. One such feature is the total number of days a student was registered for a course namely $total\_reg\_days$. This feature is derived by subtracting the $registration\_date$ from the $unregistration\_date$ in **student_registration.csv**.

Data aggregation: It helps in summarizing information and reducing the dataset's dimensionality while retaining important insights. We aggregate the VLE data by grouping it based on the $code\_module$, $code\_presentation$, $id\_student$, and $date$, then summing the total number of clicks for each group. This aggregation results in a new feature, $total\_clicks$, which represents the engagement of each student on each day. Students' interactions in the VLE are aggregated by $total\_clicks$, representing daily online engagement of students for each subject, is depicted in Figure 3. This reduces data dimensionality and creates a meaningful feature to assess online activity. We aggregate the clickstream data by grouping it based on student identifiers and course details, summing the total number of clicks each day. Engagement is assumed to be proportional to the number of interactions a student has with the course materials, forums, and other VLE activities. For example, some actions such as browsing through course material is considered strong online engagement.

Merging datasets: To facilitate analysis, we merge the **student_info.csv**, **student_registration.csv**, and **student_vle.csv** dataset files into a single dataset based on common columns ($code\_module$, $code\_presentation$, and $id\_student$). This merging combines demographic information, registration details, and engagement metrics into a single dataset. The merged dataset represents all relevant features of each student in one place, which makes it easier for us to analyze and model their performance.

Averaging Assessment: Assessment weights show the importance of each assessment in the grading of a course. The computation of the average weights helps us understand the assessment structure across different modules and presentations. First, we group the **assessments.csv** dataset file by $code\_module$ and $code\_presentation$. Second, we calculate the mean weight of the assessments within each group. This aggregation represents average weights for assessments in each module and presentation. Finally, we merge the calculated average weights with the **student_info.csv** dataset file to add this as a new feature namely *weight* in the merged dataset.

Handling missing values

The missing values can occur due to various reason, for example, error during data entry or incomplete data collection. In our dataset, we handle missing values of $unregistration\_date$ by filling them with the number of days equal to the course duration (270 days). This approach helps maintain the dataset's integrity without dropping potentially valuable rows. To handle missing $unregistration\_dates$ values, we fill them with 270 days, i.e., the course duration. This data imputation ensures that no dataset records (rows) are dropped. However, alternative methods, such as using the student's last active date can also represent the students' dropout patterns.

Encoding

In order to handle high dimensionality in huge datasets, categorical variables must be encoded correctly. Effective encoding methods, such as integer encoding, assist in lowering the dimensionality of the dataset, improving its manageability and minimizing its susceptibility to the dimensionality curse. By using integer encoding, every category in the dataset is given a distinct integer. We convert the categorical features into numerical values. To assign a unique integer to each category, integer label encoding has been performed. 0: Distinction, 1: Fail, 2: Pass, 3: Withdrawn.

Feature scaling

Normalization or scaling makes sure that all features contribute equally. This is particularly crucial for huge datasets because feature scales can differ greatly. The data are changed via standard scaling to have a standard deviation and a mean of 1 and 0, respectively. This normalization method is especially helpful for algorithms like SVM, K-NN, and neural networks that rely on the assumption that data is normally distributed or are sensitive to the size of the data. Standard scaling guarantees that, in models every feature contributes equally to the distance computations. It standardizes the range of independent variables or features by bringing all features to a similar scale, improving the model performance and training stability. In this paper, the *StandardScaler* has been considered to regulate the variables by confiscating the mean and scaling to unit variance. The z-normalization also called z-score normalization or standardization which can be mathematically represented as:

$$z = \frac{x - \mu}{\sigma} \tag{1}$$

where x, μ, σ, and z are the original value, the average (mean), the standard deviation, and the normalized value of the feature, respectively.

code_module	code_presentation	id_student	id_site	date	sum_clicks
AAA	2013J	28400	546652	-10	4
AAA	2013J	28400	546614	-10	1
AAA	2013J	28400	546876	-10	2
BBB	2014J	30268	546652	-10	3
BBB	2014J	30268	546662	-10	2
BBB	2014J	30268	546614	-10	3
BBB	2014J	30268	546652	-10	5

code_module	code_presentation	id_student	date	total_clicks
AAA	2013J	28400	-10	7
BBB	2014J	30268	-10	13

Fig. 3. Aggregation of clickstream interactions.

Class weights
Class weights are used to address class imbalance, ensuring that the model does not bias towards the majority class. While the weights can be chosen based on the inverse frequency of each class, we found the following simple class weighting scheme useful in our case: Distinction = 1.5, Fail = 1.5, Pass = 1, and Withdrawn = 1. We provide these weights to the model during training to emphasize underrepresented classes.

Building and training the 1D-CNN model

Figure 4 shows the building and training of the 1D-CNN model to estimate the performance of student. The model design consists of convolutional layers to extract the feature, fully interconnected layers for classification, and numerous techniques such as dropout and batch normalization to improve performance and prevent overfitting. Moreover, we implemented cross-validation with stratified sampling, dropout regularization, class weighting for minority categories, early stopping, and selective feature retention. Additionally, benchmarking against RF, ANN-LSTM, and DFFNN ensured the model's robustness.

- Model compilation: After defining the architecture, the sparse categorical cross-entropy loss, accuracy metric, and Adam optimizer have been used to compile the model.
- Model training: The model has been trained on the training data, specifying batch size = 1024, number of epochs = 70, and validation split = 0.1. The dataset has been split into (70%) training and (30%) testing sets. We also ensure that the split is stratified based on the target variable to maintain the distribution of classes.

The proposed model has been trained by considering the processed data, and its performance has been assessed based on F1-score, recall, precision, and accuracy. Here is a detailed explanation of the model layers.

Input layer
This layer delineates the configuration of the input data that the model will process. It anticipates each sample to possess a shape of $(X\_train.shape[1], 1)$, where the initial entry of the input information (data) signifies the number of features, which is 15 in this instance. The numeral 1 signifies that the input data is one-dimensional, such as a time series or a sequence of features. This is the initial phase of the model, ensuring that the input data is appropriately formatted. The input data can be represented as $[X \in \mathbb{R}^{15\times 1}]$.

Conv1D layer
The input data is subjected to 16 filters (small learnable weights) by the first Conv1D layer. Because each filter has a kernel size of 3, it can examine three features in succession at once. The model learns the spatial correlations between features with the aid of this layer. For example, it could discover patterns in an educational dataset connecting students' activity data over time or between various kinds of interactions (such forum postings and assignments). The activation function *ReLU* adds non-linearity to our proposed novel model, enabling it to learn intricate forms. In order to help avoid negative numbers from distorting the results, *ReLU* outputs zero if the value is not positive. If the value is positive, it outputs the value directly. The convolution operation in mathematics is expressed as;

$$Y_{1CL}[i,k] = ReLU\left(\sum_{j=0}^{3} W_{1,k,j}.X[i+j-1] + b_{1,k}\right), \tag{2}$$

where $Y_{1CL}[i,k]$ is the output for filter k at position i, $W_{1,k,j}$ are the filter weights, and $b_{1,k}$ is the bias for filter k. The shape of output after applying first 1D-convolutional operation is $Y_{1CL} \in \mathbb{R}^{15\times 16}$.

Maxpooling layer
This layer does down-sampling by selecting the maximum value within each 2-feature frame, hence halving the data's dimensionality (from 15 to 7). Max pooling decreases data complexity by emphasizing the most salient

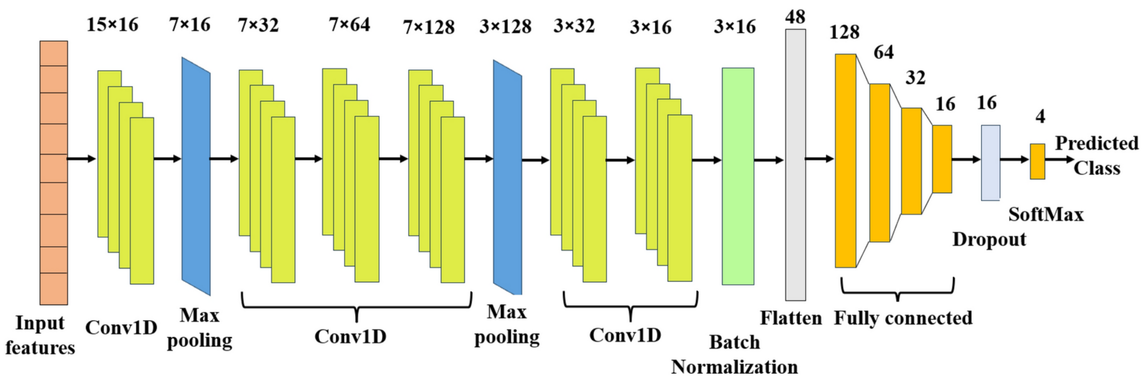

Fig. 4. 1D-CNN (Convolutional Neural Network) Model.

characteristics identified by the convolutional layer. It also aids in rendering the model invariant to minor translations of the input (e.g., modest fluctuations in feature values). It can be expressed as;

$$Z_{1MP}[i,k] = \max(Y_{1CL}[2i,k], Y_{1CL}[2i+1,k]), \quad (3)$$

where $Z_{1MP}[i,k]$ is the down-sampled output for the *ith* position and *kth* filter. The shape of data after max pooling is $Z_{1MP} \in \mathbb{R}^{7\times16}$.

Batch normalization layer
Batch normalization standardizes the outputs of preceding layers. It modifies and normalizes the activation by ensuring that the mean and the standard deviation of each batch approximates 0 and 1, respectively. This layer stabilizes the training process by mitigating significant fluctuations in data distribution as it traverses the network, hence facilitating faster convergence and enhancing generalization.

$$\hat{Y}_{BN}[i,k] = \frac{Y_{6CL}[i,k] - \mu k}{\sqrt{\sigma_k^2 + \epsilon}} \quad (4)$$

where μ_k, σ_k^2, and ε is the mean of *kth* feature map, variance of *kth* feature map, and the small constant added for numerical stability, respectively.

The multi-dimensional data from the convolutional layers is flattened into a 1D vector by the flatten layer. To feed the data into the dense (fully connected) layers, it must first be flattened. It converts the three-dimensions (3D) tensor (batch size, feature-length, number of filters) into a 2D tensor (batch size, total features).

$$Y_{FL} = \zeta(\hat{Y}_{BN}) \quad (5)$$

where ς represents the flatten operation, $\hat{Y}_{BN}$ is the output of the batch normalization, and Y_{FL} is the output of flatten layer.

Fully connected layers
There are 128 neurons in the initial completely connected (dense) layer. Every neuron is entirely linked to all outputs from the preceding layer. Dense layers execute the classification operation by analyzing the characteristics derived from the convolutional layers. The *ReLU* activation facilitates the acquisition of intricate correlations among the characteristics. Mathematically represented as;

$$Z_{1DL} = ReLU(W_{1DL}.Y_{FL} + b_{1DL}) \quad (6)$$

$Z_{1DL} \in \mathbb{R}^{128}$ represents the output dimension of the initial fully connected layer. The second dense layer comprises 64 neurons, which further distills the information, enabling the model to concentrate on the most salient aspects. The operation of Equation (6) has been executed for the second, third, and fourth dense layers. The data size following the fourth dense layer is $Z_{4DL} \in \mathbb{R}^{16}$.

Dropout layer
30% of the input units are randomly set to 0 by the dropout layer during each step of training, denoted by $Z_{drop} = (Z_{4DL}, \rho = 0.3)$. By keeping the model from being overly reliant on any one neuron, this keeps it from overfitting. Dropout reduces the likelihood of overfitting the training set and increases the capacity of model for generalization.

Output layer
There is $n\_classes = 4$ units in the output layer, one for each class. The activation function Softmax has been consider to create a probability distribution over the different categories, where each unit represents the likelihood which the input belongs to a certain category. Multiclass classification issues are solved using Softmax, which is expressed as $Z_{final} = soft\max(Z_{drop})$. Data $Z_{final} \in \mathbb{R}^4$ is the size of the data. This layer outputs the probability distribution among the classes. This layer outputs the probability distribution among the classes. The reason for using sparse categorical cross-entropy is that the labels are integers rather than one-hot encoded. For multi-class classification problems, this loss function is suitable, as it is described as,

$$loss = -\frac{1}{N}\sum_{i=1}^{N} \log(P(y_i|X_i)), \quad (7)$$

where *N* denotes the number of samples and $(P(y_i|X_i))$ represents the projected probability for the true class y_i given the input X_i. Adam serves as the optimizer. Adam is an optimization method with an adaptive learning rate that has gained significant popularity owing to its efficiency and performance.

The 1D-CNN model analyses sequential data by employing convolutional layers for feature extraction, down-sampling via max pooling, and normalizing through batch normalization. The extracted features are flattened and fed to fully connected layers. To mitigate overfitting, dropout layers are also employed in this part of the network. Finally, the output passes through a Softmax function to generate a probability distribution for the four classes. Our proposed model employs sparse categorical cross-entropy loss function and the Adam optimizer.

The 1D-CNN model efficiently captures sequential patterns in student interactions within a VLE to identify temporal dependencies in their behavior. Its computational efficiency makes it ideal for real-time predictions. Since the student engagement data is a time-series, 1D-CNN effectively detects early signs of disengagement or success.

The 1D-CNN efficiently extracts features from high-dimensional clickstream data by capturing local patterns. It is computationally scalable because it can handle large datasets (30,000+ students) more efficiently than models like LSTM. Its strong performance in multiclass classification enables accurate predictions of student outcomes (Distinction, Pass, Fail, Withdrawn), even when early behavioral data is sparse. This makes the 1D-CNN a robust choice for modeling student engagement and performance trends.

Experimental outcomes and discussion on results

The early prediction, the experimental setup as well as the experimental results are discussed in this section. The pre-processed dataset, detailed in Section 3, has been used for analysis.

Experimental setup

The proposed method was evaluated by comparing it with various baseline models using different datasets and feature sets. First, it was compared with Random Forest (RF) models trained with gini index (RF 'gini'[24]) and entropy (RF 'entropy'[24]) using all features. Next, the proposed method was compared to DFFNN[4]and ANN-LSTM[15]using demographic and clickstream data, as well as to DFFNN[4] with demographic, clickstream, and assessment data. Early forecasting experiments were conducted with different training dataset sizes (5%, 10% , 20%, and 100%) to assess training accuracy and loss vs validation accuracy and loss. Performance metrics, including precision, recall, F1-score, and accuracy, have been evaluated at different course lengths (5%, 10%, 20%, 40%, 60%, 80%, and 100%). Finally, ROC curves and confusion matrices were generated for the proposed method on the test set at varying course lengths to assess classification performance. All models are implemented and the experiments are run using TensorFlow and Keras.

Experimental observations and test results

Table 2provides a comparative analysis between the effectiveness of the suggested methodology against RFs models utilizing 'gini' and 'entropy' criteria[24], and DFFNN[4] trained and evaluated on the comprehensive OULA dataset for multiclass classification (Pass, Distinction, Withdrawn, and Fail categories). The assessment measures encompass F1-score, recall, precision, and accuracy, for all four categories. The suggested strategy regularly surpasses the baseline RF with different criteria such as 'gini' and 'entropy' models, and also DFFNN, yielding improved outcomes across all metrics. This indicates that the proposed method is superior in accurately identifying student outcomes, which results in more robust predictions.

Table 3provides a comparison of the proposed method with the ANN-LSTM[15]and DFFNN[4] based methods, when we utilize demographic and clickstream variables for multiclass classification (Pass, Distinction, Withdrawn, and Fail categories). The proposed methods exhibits superior performance in terms of accuracy, recall, precision, and F1-score. The higher recall values indicate that the proposed method is able to identify most true occurrences of each class with a small number of false negatives. The precision and F1-score for the proposed method is also higher than other methods.

Table 4shows a comparison of the proposed method with DFFNN[4], when clickstream, assessment, and demographic features are used. For all the four classes, the results are displayed in terms of macro average, weighted average, F1-score, recall, precision and accuracy. The proposed method consistently performs better than the DFFNN in Table 4, especially when more features are taken into account. The improved F1-score, macro, and weighted averages demonstrate the robustness of the proposed approach and demonstrate its capacity to generate accurate forecasts across all four classes.

The proposed 1D-CNN model outperforms RF, DFFNN, and ANN-LSTM in terms of accuracy, precision, recall, and F1-score. Its ability to capture sequential dependencies allows it to recognize early disengagement patterns, unlike RF and DFFNN, which lack temporal awareness. Effective feature engineering, such as aggregating VLE interactions, further enhances its predictive performance compared to ANN-LSTM.

	Proposed			RF 'gini'[24]			RF 'entropy'[24]			DFFNN[4]		
	Precision	Recall	F1-score	Precision	Recall	F1-score	Precision	Recall	F1-score	Precision	Recall	F1-score
Distinction	**0.97**	**0.99**	**0.98**	0.66	0.48	0.56	0.66	0.48	0.56	0.64	0.49	0.55
Fail	**0.97**	**0.96**	**0.96**	0.56	0.45	0.50	0.55	0.44	0.49	0.55	0.37	0.44
Pass	**0.99**	**0.99**	**0.99**	0.69	0.86	0.77	0.69	0.87	0.77	0.76	0.91	0.82
Withdrawn	**1.00**	**0.99**	**0.99**	0.64	0.50	0.57	0.65	0.48	0.56	0.76	0.80	0.78
Accuracy	**0.98**			0.66			0.66			0.72		
Macro average	**0.98**	**0.98**	**0.98**	0.64	0.57	0.60	0.64	0.57	0.59	0.68	0.64	0.65
Weighted average	**0.98**	**0.98**	**0.98**	0.65	0.66	0.65	0.65	0.66	0.64	0.70	0.72	0.70

Table 2. Comparison of the proposed method with RF 'gini'[24], RF 'entropy'[24], and DFFNN[4], when the training data includes all features.

	Proposed			ANN-LSTM[15]			DFFNN[4]		
	Precision	Recall	F1-score	Precision	Recall	F1-score	Precision	Recall	F1-score
Distinction	**0.92**	**0.97**	**0.94**	0.82	0.47	0.59	0.77	0.02	0.04
Fail	**0.88**	**0.90**	**0.89**	0.79	0.55	0.65	0.52	0.20	0.29
Pass	**0.96**	**0.96**	**0.96**	0.84	0.60	0.70	0.62	0.92	0.74
Withdrawn	**0.93**	**0.80**	**0.86**	0.82	0.62	0.70	0.67	0.76	0.71
Accuracy	**0.94**			0.72			0.63		
Macro average	**0.92**	**0.91**	**0.91**	0.82	0.56	0.66	0.64	0.48	0.45
Weighted average	**0.94**	**0.94**	**0.94**	0.82	0.58	0.68	0.63	0.63	0.56

Table 3. Comparison of the proposed method with DFFNN[4]and ANN-LSTM[15], when clickstream data and demographic information is used to predict student performance. The Precision, Recall, and F1-score are provided for each class. The Macro average calculates the mean performance across all classes without considering class size, i.e., equal weighting for each class. The Weighted average adjusts for class imbalances by weighting the performance of each class according to its frequency in the dataset. This metric is crucial when dealing with imbalanced classes.

	Proposed			DFFNN[4]		
	Precision	Recall	F1-score	Precision	Recall	F1-score
Distinction	**0.91**	**0.98**	**0.95**	0.66	0.47	0.55
Fail	**0.92**	**0.90**	**0.91**	0.55	0.36	0.44
Pass	**0.96**	**0.97**	**0.97**	0.74	0.90	0.81
Withdrawn	**0.95**	**0.83**	**0.89**	0.76	0.80	0.78
Accuracy	**0.95**			0.71		
Macro average	**0.94**	**0.92**	**0.93**	0.68	0.63	0.64
Weighted average	**0.95**	**0.95**	**0.95**	0.70	0.71	0.70

Table 4. Comparative analyses of the proposed method and DFFNN[4] models utilizing demographic, assessment, and clickstream data.

Furthermore, the use of class weights in 1D-CNN model handles class imbalances well to distinguish categories with small number of samples such as Withdrawn and Distinction better than RF and DFFNN. Despite these strengths, the model is prone to overfitting when trained on smaller datasets. Furthermore, its performance may be sensitive to choice of hyperparameter, that can impact model generalizability across different datasets. In comparative performance, RF and DFFNN handle class imbalances well but fail to capture sequential data patterns. ANN-LSTM, while designed for sequential learning, struggles with noise in engagement data and performs less effectively than the 1D-CNN.

Figure 5 depicts the influence of several factors on the final output. The $total\_reg\_days$ is identified as the most relevant feature, with a significance value of 0.30. Other features also influence performance, including $studied\_credits$, $num\_of\_prev\_attempts$, imd_band, and $highest\_education$, among others. The $total\_reg\_days$ feature is derived through feature engineering from the reg_date and $unreg\_date$. Only the $unreg\_date$ for the Withdrawn class is accessible in the OULA dataset. The $unreg\_date$ is crucial for other classes for differentiation, as evidenced by the value of total reg days illustrated in the Figure 5. Figure 6 shows the effect of feature $total\_reg\_days$ on the Final Results of the students, which is created by using feature engineering on the features reg_date and $unreg\_date$ (only mentioned of Withdrawn class in the dataset). This feature clearly represents the discrimination in all classes including Distinction, Fail, Pass, and Withdrawn.

Early prediction

Figure 7 compares training and validation accuracy as well as loss to show how varying course length percentages (5%, 10%, 20%, and 100%) affect model performance. The model exhibits better generalization as the dataset size grows, with a steadier validation loss and a lower difference between training and validation accuracy. Significant overfitting problem transpires when the model performs poor on validation data but better on training data, and this is the case with smaller datasets (e.g., 5%). The training and validation measures, on the other hand, more closely correlate with 100% of the dataset, suggesting that the model benefits from more data and is less prone to overfitting. This illustrates how higher model performance and generalization can be attained with larger datasets. Even when trained at a small dataset size of 20%, our suggested model achieves improved training loss and accuracy as well as validation loss and accuracy. When the data size increases from 5% to 10% and 20%, the model flawlessly aids in the early prediction of students who are at risk.

Figure 8 shows the model performance of F1-score, recall, precision, and accuracy while considering the 5% , 10%, 20%, 40%, 60%, 80% and 100% course duration. The results of classes: Distinction, Fail, and Pass are

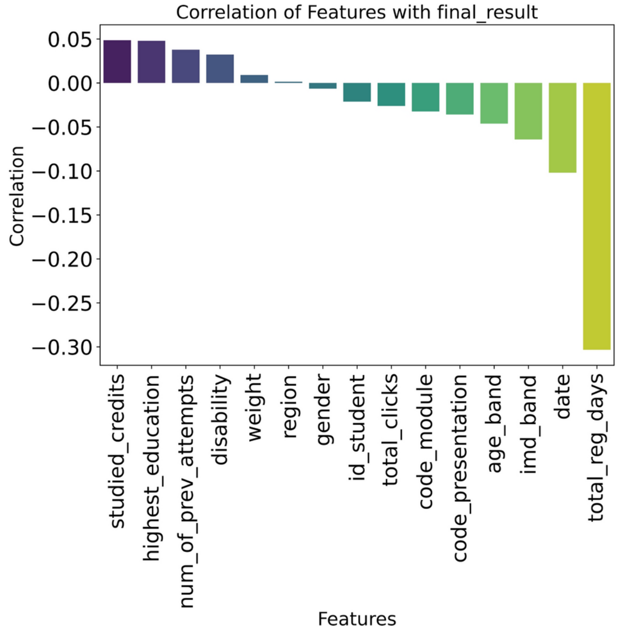

Fig. 5. Correlation analysis of features with final_result.

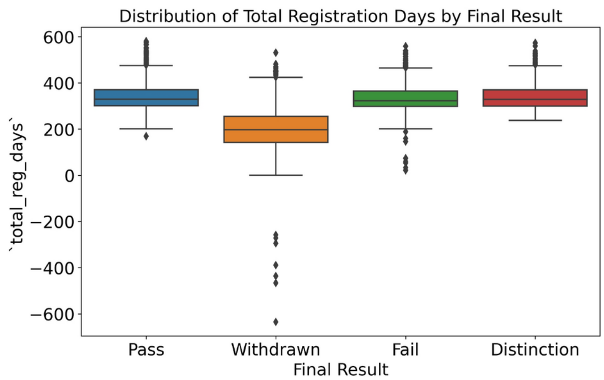

Fig. 6. Class distribution against $total\_reg\_days$.

more than 90% in terms of F1-score, recall, precision, and accuracy as the course duration increases more than 20%. While the Withdrawn class performance is much better than other classes even at 5% of course duration. The reason is that the reg_date and $unreg\_date$ of Withdrawn students are mentioned but for other classes students only reg_date is available in the OULA dataset. Thus, we have created a new feature $total\_reg\_days$ = reg_date - $unreg\_date$ where a value of 270 has been considered for classes whose $unreg\_date$ is not available in the dataset because of the last day of course duration. At the 5% course duration, the model demonstrates strong early performance for the Withdrawn category due to the inclusion of registration and unregistration data. However, for other categories such as Distinction, Fail, and Pass, the predictions are less accurate, as the model has fewer interactions and engagement data to rely on. Distinction predictions improve later as the model captures more engagement signals, while Fail predictions benefit from earlier interactions once students begin to disengage from the course. The Pass category shows steady improvement as the course progresses, with the model relying on early engagement as well as assessment data.

Taking into account all relevant features, Table 5 shows the early prediction students' performance in VLE at different points in the course progression. Metrics like F1-score, recall, precision, and accuracy has been considered for the performance assessment at various prediction intervals including 5%, 10%, 20%, 40%, 60%, and 100% of the course duration. The following categories are evaluated: Withdrawn, Distinction, Fail, and Pass. When it comes to Distinction, the model performs moderately at first (F1-score: 0.75, Recall: 0.74, Precision: 0.76) at 5% of course duration, but by the end of the course, it has much improved (0.97, 0.99, 0.98), demonstrating high accuracy to detect the students those are expected to excel. For Fail class, predictions start off with lower scores (0.70, 0.63, 0.66) at 5% of course duration, but these rise to 0.97, 0.96, and 0.96 by considering 100% of course duration, demonstrating that more data leads to a better identification of failed students. For Pass,

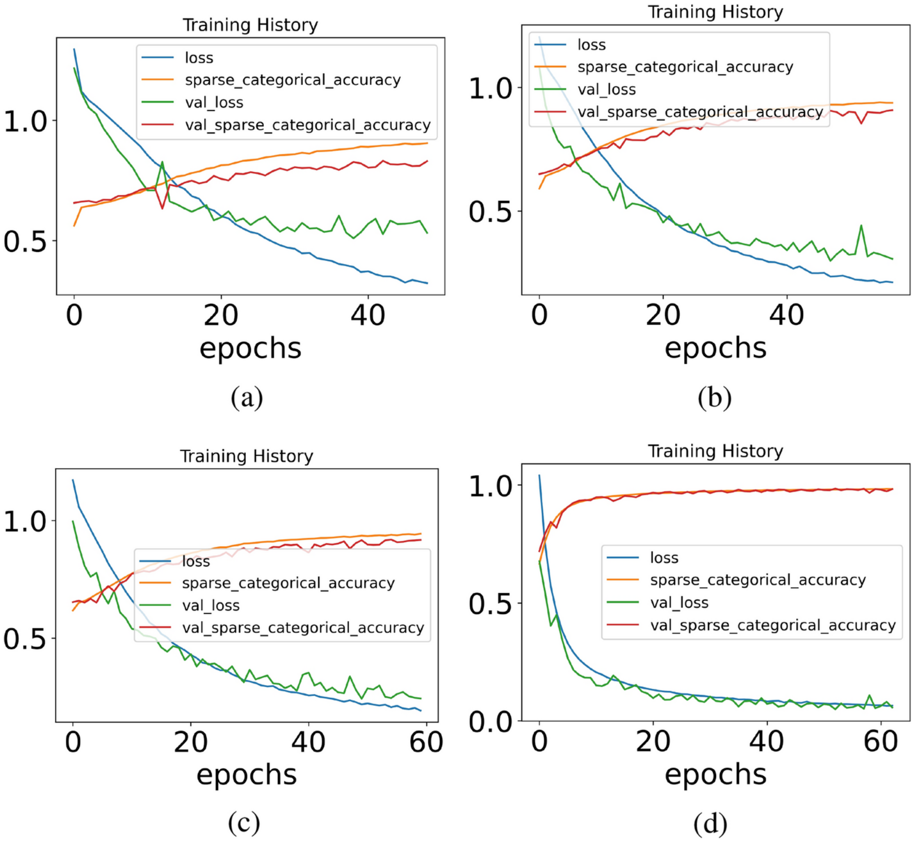

Fig. 7. Training vs validation accuracy and loss at different training of dataset size (**a**) 5%, (**b**) 10%, (**c**) 20% and (**d**) 100%.

the model shows dependable forecasts of passing students early on (0.82, 0.85, 0.84) and reaches near-perfect accuracy (0.99 for all metrics) at 100% of course length. Withdrawn exhibits the best overall performance, attaining near-perfect scores by 100% course duration (1.00, 0.99, 0.99) after beginning high at 5% (0.96, 0.95, 0.95). This indicates the model's resilience in predicting students those are expected to withdraw. The model's overall accuracy increases from 0.82 to 0.98 while course duration is considered 5% to 100%, highlighting the growing dependability of predictions with additional course data, especially for the Pass and Withdrawn categories where scores regularly approach or surpass ideal levels.

ROC with different course length

Figure 9 depicts the ROC curve performance of a multiclass classification model across four classes: Distinction, Pass, Fail, and Withdrawn with different percentages of course duration (5%, 10%, 20%, and 100%). Each solid line represents the ROC curve for one class, showing how well the model distinguishes that class from the others. The micro-average and macro-average curves are represented by dotted lines (pink and blue), which provide overall performance metrics, with AUC values of 0.9586 and 0.9424 while considering the 5% of course duration, respectively, indicating strong overall classification ability as presented in Figure 9 (a). The individual AUC values for the classes range from 0.9059 to 0.9878, with the Withdrawn class showing near-perfect discrimination, highlighting the model's robustness in distinguishing between the different categories. As represented in Figure 9 (c), for the 20% of course length, the AUC values for micro-average and macro average curves are 0.9904 and 0.9870, respectively, which can be considered as a perfect early prediction of students' performance. Even the AUC curves values for Distinction, Fail, Pass, and Withdrawn are 0.9922, 0.9773, 0.9806, and 0.9979, respectively. As we consider the course length more than 20%, the model performance becomes accurate (almost 100%) in terms of individual AUC values for the classes as well as micro and macro-average, Figure 9. The ROC curves for our model indicate high AUC values, which validate the model's strong ability to distinguish between classes at different stages of the course. For example, at the 5% and 10% marks, the AUC values for the Withdrawn category are particularly high, indicating the model's capacity to identify students at risk of withdrawal. In contrast, for the Distinction and Fail categories, the AUC values improve significantly as the course progresses, which implies that the model requires more data to accurately predict these classes. Compared to existing studies, our model performs well in the early stages (such as 5% to 20% course duration) that typically focus on the later stages of the course. The results show the effectiveness of the proposed model in an online education environment for accurate early at-risk prediction of student performance that will be very helpful for an intelligent education system.

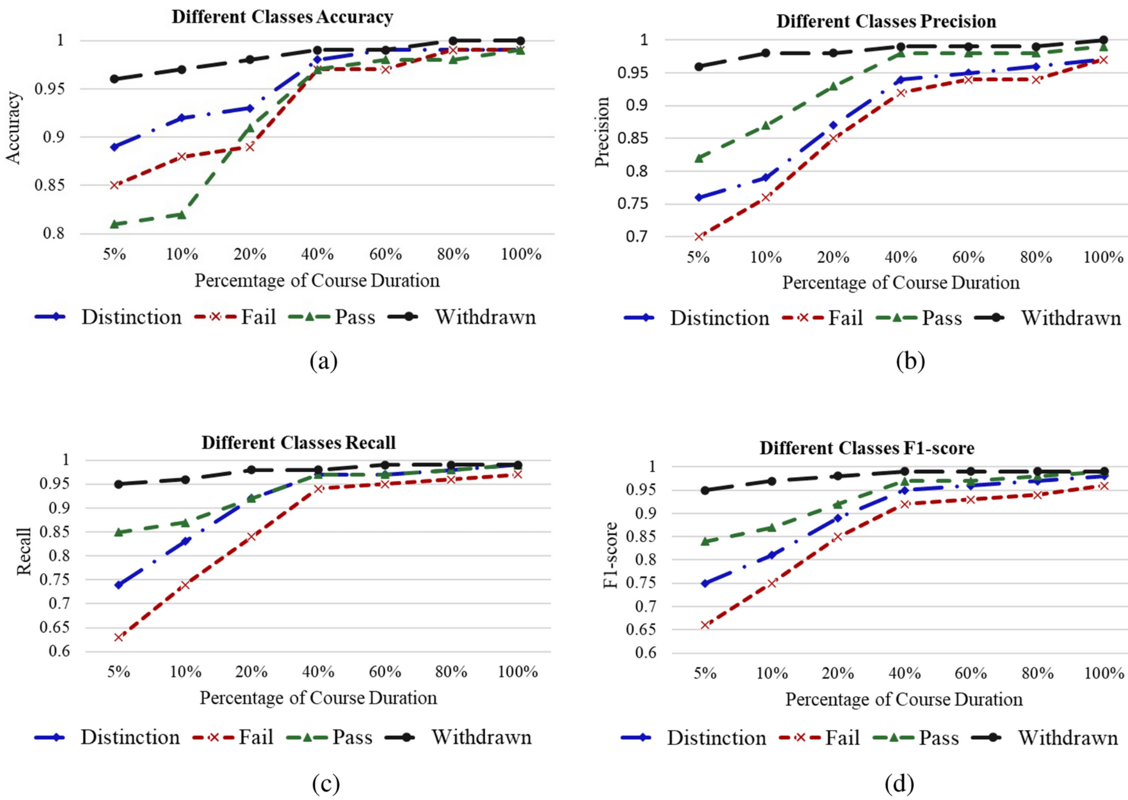

Fig. 8. Performance of different classes with the percentage of course length (**a**) Accuracy, (**b**) Precision, (**c**) Recall and (**d**) F1-score.

	5%course duration			**10%course duration**			**20%course duration**		
	Precision	**Recall**	**F1-score**	**Precision**	**Recall**	**F1-score**	**Precision**	**Recall**	**F1-score**
Distinction	0.76	0.74	0.75	0.79	0.83	0.81	0.87	0.92	0.89
Fail	0.70	0.63	0.66	0.76	0.74	0.75	0.85	0.84	0.85
Pass	0.82	0.85	0.84	0.87	0.87	0.87	0.93	0.92	0.92
Withdrawn	0.96	0.95	0.95	0.98	0.96	0.97	0.98	0.98	0.98
Accuracy	0.82			0.86			0.92		
Macro average	0.81	0.79	0.80	0.85	0.85	0.85	0.91	0.91	0.91
Weighted average	0.82	0.82	0.82	0.86	0.86	0.86	0.92	0.92	0.92
	40%course duration			**60%course duration**			**100%course duration**		
	Precision	**Recall**	**F1-score**	**Precision**	**Recall**	**F1-score**	**Precision**	**Recall**	**F1-score**
Distinction	0.94	0.97	0.95	0.95	0.97	0.96	**0.97**	**0.99**	**0.98**
Fail	0.94	0.94	0.94	0.92	0.95	0.93	**0.97**	**0.96**	**0.96**
Pass	0.98	0.97	0.97	0.98	0.97	0.97	**0.99**	**0.99**	**0.99**
Withdrawn	0.99	0.98	**0.99**	**1.00**	**0.99**	**0.99**	**1.00**	**0.99**	**0.99**
Accuracy	0.97			0.97			**0.98**		
Macro average	0.96	0.96	0.96	0.96	0.97	0.96	**0.98**	**0.98**	**0.98**
Weighted average	0.97	0.97	0.97	0.97	0.97	0.97	**0.98**	**0.98**	**0.98**

Table 5. Early prediction of students' performance in online learning while considering all features.

Confusion matrix with different course length

Figure 10 presents confusion matrices with course duration of 5%, 10%, 20%, and 100% while all features have been considered for training and testing process of the model. As the course length increases from 5%, 10% , 20%, and 100%, the confusion matrix values in diagonals improve for all categories. The Withdrawn class value is exactly 0.96, 0.98, and 0.99 while course duration is 10%, 20%, and 100% as presented in Figure 10 (b), (c), and (d). While the diagonal values of confusion matrices for other classes including Distinction, Fail, and Pass are almost the same at 20% or more course lengths as depicted in Figure 10 (c) and (d). The confusion

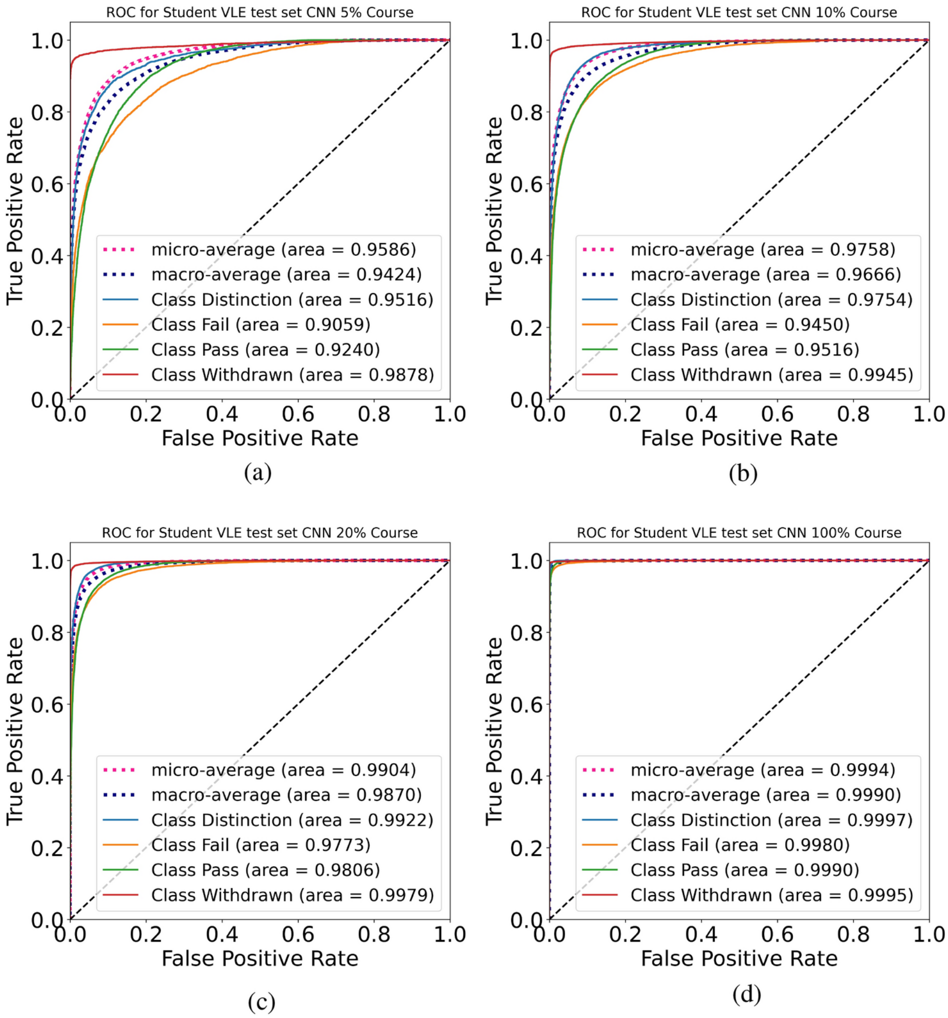

Fig 9. Test set ROC for the proposed 1D-CNN at different course lengths: (**a**) 5% (**b**) 10% (**c**) 20%, and (**f**) 100%.

matrices reveal that the Withdrawn category is predicted with high accuracy, with minimal false positives and false negatives, especially at early stages of the course. For Distinction and Fail, the model tends to misclassify some students as Pass, particularly in the early stages, where engagement data is limited. The Pass category shows fewer misclassifications overall but still experiences some false negatives when students fail to engage early on. These errors can be attributed to the model's reliance on early-stage interaction data, which may not fully capture a student's potential until later in the course. The results suggested the proposed idea is suited to the VLE for the early at-risk performance prediction of the students.

Model sensitivity

Figure 11 presents a comprehensive sensitivity analysis of key hyperparameters including Dropout rate, Kernel size, Batch size, and Learning rate, illustrating their impact on model performance in terms of classification accuracy. Figure 11 (a) shows how different dropout rates: 0.1, 0.2, 0.3, 0.4 influence the accuracy. Dropout is a regularization technique used to prevent overfitting by randomly dropping a fraction of the neurons during training. From the graph, it can be observed that moderate dropout rates (0.1, 0.2, and 0.3) lead to relatively higher accuracy, suggesting that they may offer a good balance between regularization and model performance. A dropout rate of 0.4, however, shows slightly lower accuracy, indicating over-regularization, which might hinder the model from learning effectively. Figure 11 (b) presents the effect of varying kernel sizes: 3x3, 5x5, 7x7, and 9x9 on model performance. Smaller kernel sizes like 3x3 exhibit higher accuracy than larger kernels (7x7 and 9x9). This can imply that larger kernels may cause the model to lose valuable details. Figure 11 (c) examines the influence of different batch sizes (256, 1024, 4096, 16384) on the model's accuracy. it is obvious to note that a batch size of 1024 performs better than others. Figure 11 (d) plots accuracy of the model against different learning rates: 0.01, 0.005, 0.001, and 0.0005. It can be seen that a moderate learning rate of 0.001 yields the best accuracy, while very high (0.01) or very low (0.0005) learning rates result in lower accuracy.

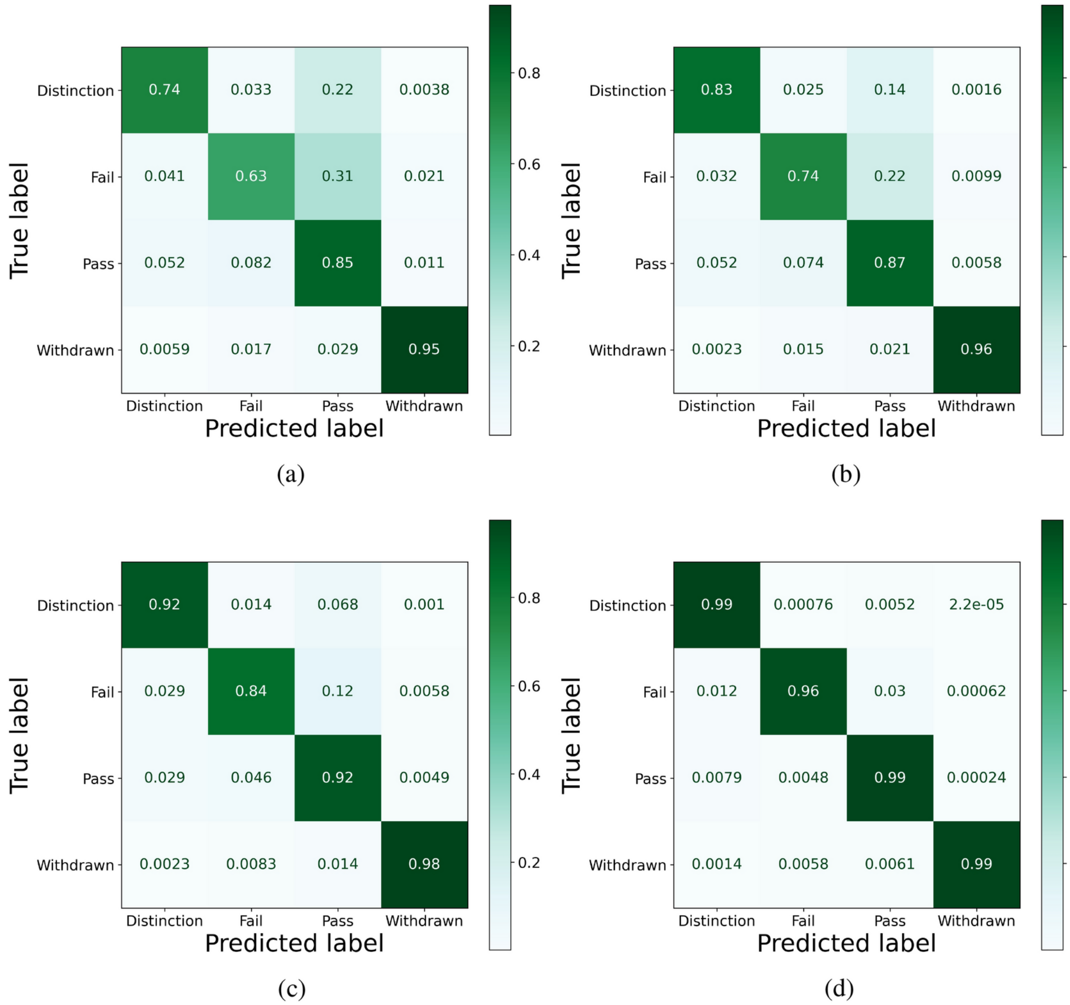

Fig. 10. Test set confusion matrices for the proposed 1D-CNN at different course lengths considering all features: (**a**) 5%, (**b**) 10%, (**c**) 20% and (**d**) 100%.

Conclusion and future work

This paper proposed a novel method to predict early at-risk students in a VLE. The method utilized a novel feature engineering approach and a 1D-CNN model to assess student performance using the OULA dataset. The model showed significant improvements in terms of accuracy, precision, recall, and F1-score over baseline methods: RF 'gini', RF 'entropy', DFFNN, and ANN-LSTM. The method demonstrated excellent predictive power at early stages of the online courses, with promising results at 20% and 40% of the course duration. At the full course duration of 100%, with all features considered, the model accurately identifys at-risk students by attaining an average accuracy and F1-score 98%. The results suggest that our method can enhance early intervention strategies in online education platforms to enable educators to address students' needs.

However, the model's performance can vary in other online educational platforms due to differences in course structures, student demographics, and the quality of the available data. Particularly, in contrast to the OULA dataset, not all educational platforms provide rich clickstream data and interaction logs. These factor can affect the model's ability to accurately predict student outcomes. Moreover, the model's reliance on course duration may not generalize to courses with different lengths or formats.

While the proposed method performed well on the OULA dataset (data from a single institution); however, its performance on other datasets have yet to be determined. To evaluate its robustness across various environments, future research will focus on adapting and testing the model in other educational settings and online platforms. We plan to explore the integration of other types of data, such as interactive learning tools and student feedback, which could enhance the model's performance. Furthermore, we aim to examine Transformer-based models, which have shown promise in other domains such as natural language processing and time-series analysis. These future works will contribute to develop predictive models that can make early interventions in online education more effective.

Finally, future research will explore how different types of data, such as demographic or engagement metrics, can be utilized in other educational settings. This will ensure the model's adaptability and help to understand the potential biases introduced by relying on a single dataset.

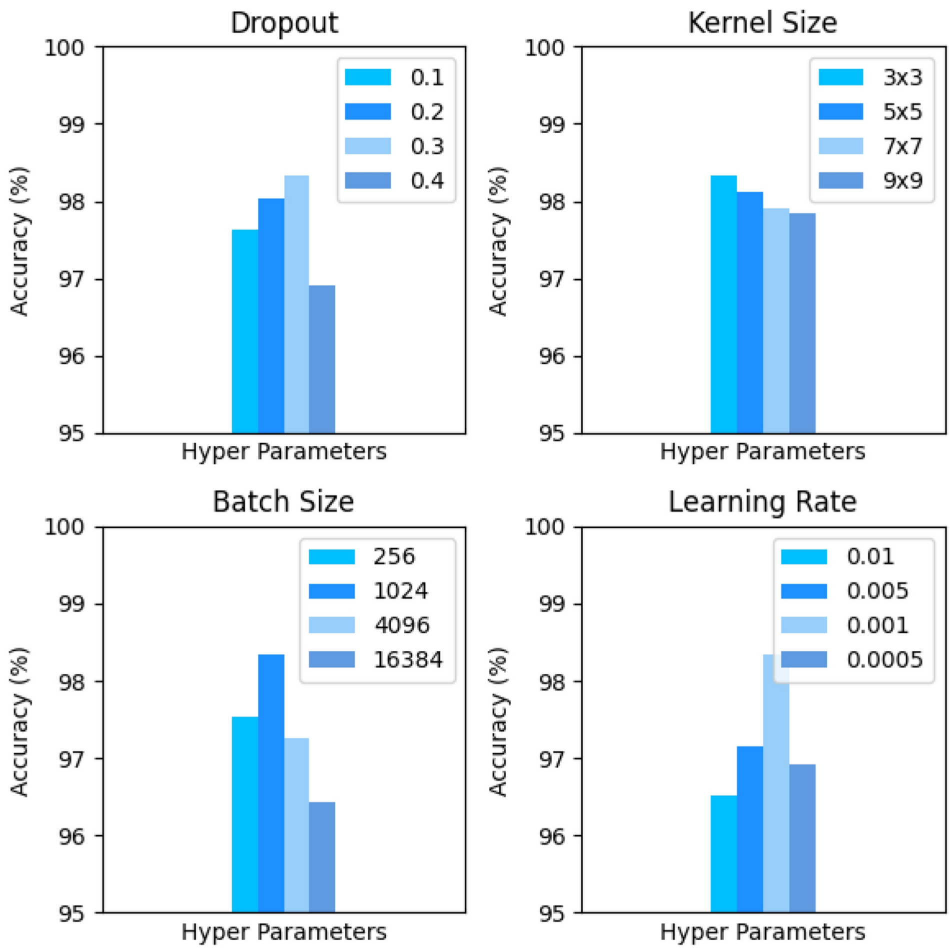

Fig. 11. Model sensitivity for various values of hyperparameters: dropout, kernel size, batch size, and learning rate.

Data availability

The dataset analysis for the experiments is publicly available on the website of the UK university: https://analyse.kmi.open.ac.uk/open-dataset.

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Author contributions

Investigation, Naveed Ur Rehman Junejo; Conceptualization and methodology, Naveed Ur Rehman, Muhammad Wasim Nawaz; Writing original draft, Naveed Ur Rehman Junejo, Muhammad Wasim Nawaz; Review and Editing Naveed Ur Rehman Junejo, Qingsheng Huang, Chang Wang, Gengzhong Zheng; Funding acquisition, Xiaoqing Dong, Gengzhong Zheng; Resources, Xiaoqing Dong, Gengzhong Zheng; Supervision, Qingsheng Huang, Gengzhong Zheng.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to Q.H. or G.Z.

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